

Jae Hyeon Kim

Atlanta, GA | **Email:** jkim3191@gatech.edu | **Mobile:** (404) 796 2576

LinkedIn: www.linkedin.com/in/jae-hyeon-kim-669810131 | **Website:** <https://jkim3191.github.io/>

RESEARCH INTERESTS

- Locomotion, Control, Motion & Path Planning
- Navigation, SLAM

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Robotics (Mechanical Engineering)

Expected Dec 2027 | GPA: 4.0 / 4.0

Bachelor of Science in Mechanical Engineering

Dec 2025 | GPA: 3.8 / 4.0

RESEARCH EXPERIENCE

Georgia Institute of Technology | Atlanta, GA

Laboratory for Intelligent Decision and Autonomous Robots (LIDAR Lab)

- **Project: Learning-Based Interaction-Aware Navigation (Unitree G1)** *Aug 2025 – May 2026*
 - Developing a **reinforcement learning** pipeline for bipedal **locomotion** and contact-rich behaviors, enabling stable balance during high-impulse interactions
 - Constructing vision-based decision-making model through PPO, enabling the robot to adjust its pose from RGB-D and interact with objects along its path.
 - Developed a **low-level reinforcement learning policy** for executing high-level trajectories Isaac Lab
- **Project: Bipedal Robot Traversability (Agility Robotics Digit)** *Dec 2024 – May 2025*
 - Contributed to terrain-aware **navigation** framework by developing testing environments in MuJoCo featuring uneven and cluttered terrains to stress-test bipedal locomotion stability
 - Enhanced obstacle perception modules to convert depth data into traversability constraints, supporting **self-supervised learning-based** estimation integrated with MPC and RRT* planning
 - Conducted **hardware** experiments on the Digit humanoid robot, diagnosing 'sim-to-real' discrepancies and implementing debugging solutions for system integration, sensor fusion, and state estimation refinement
 - Validated the system's robustness through extensive data collection and analysis, achieving an 80% success rate on challenging terrains and outperforming previous baselines

The Safe Robotics Lab (SRL)

- **Project: 3D Scene Generation for Autonomous Navigation** *Sep 2025 – Dec 2025*
 - Architected multi-agent **LLM** pipeline with visual feedback loops for **automated 3D scene generation** from natural language, using Pydantic AI + Blender integration
 - Developed semantic search system with dual embeddings (CLIP + SBERT) querying 18K+ assets via PostgreSQL vector operations
 - Implemented VLM-based (Gemini 2.5 Pro) iterative feedback evaluating rendered scenes to refine object placement and ensure spatial feasibility
 - Built end-to-end Blender automation exporting simulation-ready environments for **autonomous robot** testing
 - Generated 3D indoor scenes for **navigation** training without manual intervention

Undergraduate Capstone Project (Industry-Sponsored), Aetos Imaging

- **Project: Autonomous Navigation Integration for Unitree Go2** *Sep 2025 – Dec 2025*
 - Built **autonomous navigation** functionality for the Unitree Go2 by extending the Unitree Go2 SDK and integrating robot command and state feedback into ROS 2
 - Implemented control interfaces using ROS 2 publish/subscribe and services/actions for navigation commands and system status reporting
 - Added runtime safety mechanisms, including emergency stop handling, command limits, and state validation checks
 - Successfully deployed navigation stack for real-world facility **mapping**, enabling the Unitree Go2 to navigate industrial environments while recording high-fidelity digital twin data

INDUSTRY EXPERIENCE

Samsung Electronics DS | Suwon, South Korea

Wafer-Level Packaging Engineering Intern

Jun 2024 – Aug 2024

- Analyzed fabrication data in FOWLP processes to improve chip yield by identifying defects (nodule, EMC delamination)
- Investigated glass wafer cracking using mechanical analysis with simulation, SEM, and FTIR

SKILLS

- **Robotics & Control:** SLAM, MPC, State Estimation, Path Planning, Robot Dynamics
- **AI & Machine Learning:** Deep Learning (PyTorch), Computer Vision, Vision-Language Models (CLIP, Gemini), LLM-based pipelines
- **Programming:** Python, C/C++, MATLAB
- **Frameworks:** ROS 1/2, IsaacSim, MuJoCo, Gazebo
- **Tools:** Git, Docker, Linux, RViz

ACTIVITIES

- **Volunteer, Georgia Tech Science & Engineering Day (Atlanta Science Festival) | Atlanta, GA** *Mar 2024*
 - Assisted with hands-on robotics demonstrations for public STEM outreach, including a DIY robotic hand activity

PRESENTATIONS

- **Poster Presentation and Technical Demonstration, Bitsensing Inc. | Atlanta, GA** *Aug 2025*
 - Presented AI-Driven Intelligent Transportation system architecture and real-world deployment results through an interactive booth demo at the ITS World Congress

TEACHING EXPERIENCE

- **Teaching Assistant for Georgia Tech course “ME 3017: System Dynamics” | Atlanta, GA** *Fall 2025*